Tyler Wiggins

tyler.wiggins2006@gmail.com | (423) 320-6889 | linkedin.com/in/tylerhwiggins | Eligible for Security Clearance

Education

University of Tennessee — Knoxville, TN August 2025 – May 2029
B.S. Aerospace Engineering GPA: 4.00/4.00

Experience

Subject Matter Expert, Student Space Technology Association — University of Tennessee August 2025 – Present

- Provided technical guidance and design feedback on electronics projects and rocketry subsystems
- Co-led a 16-person team developing a command-and-control system for a liquid-fueled rocket
- Presented at preliminary and critical design reviews; helped implement club-wide systems engineering
- Managed version control and code reviews as a member of the club's software development team

IT and AV Intern, Firecrown Media Inc. — Chattanooga, TN June – August 2026

- Managed employee database access and gave technical support across company IT and AV infrastructure
- Coordinated with the media and events team to deploy an event app for 1,000+ attendees at FreightWaves F3

Projects

Big Liquid Command and Control Team — SSTA Aug. 2025 – Present

- Co-led development of the command-and-control platform for SSTA's 70k+ ft apogee liquid rocket
- Developed data acquisition (DAQ) system infrastructure controlling 8 sensors and 6 triggers
- Led recovery and analysis of thrust and pressure data to characterize engine performance
- Helped develop test-fire and launch procedures, enabling 5+ successful test fires
- Led systems engineering design processes for the development of mission-critical subsystems

Orbital Trajectory and Flight Control Sim — Personal Project Apr. 2026 – Present

- Built a scalable, general-purpose 6-DoF rocket guidance and dynamics simulator in C++
- Implemented RK4 integration with J2 gravity perturbations, passes all NASA NESC test cases
- Developed a staged Lambert-based guidance system to guide a rocket between two points on Earth
- Selected by SSTA for verification and validation of its thrust vector control (TVC) system

GLASS | Hardware-in-the-loop Rocketry Test Replay System — SSTA May 2026 – Present

- Designed a 3-board HIL PCB assembly that replays test-fire sensor data into the DAQ for mission simulation
- Developed an 8-channel DAC-based analog circuit to emulate load cell and pressure transducer outputs
- Achieved voltage reproduction within 0.02% error while processing data at 50+ Hz
- Led the GLASS team of 10 people building a software platform for DAQ testing and verification

PHASE Flight Controller — SSTA Jan – May 2026

- Contributed to firmware design for a flight telemetry device for a 70k+ ft apogee rocket
- Implemented a custom LoRa telemetry protocol for bidirectional flight computer/ground station communication
- Created a data collection and digital twin system for real-time flight prediction and analysis

Skills

- Programming Languages: C, C++, Python, MATLAB
- Technical: SolidWorks, Git, Linux, KiCad, Embedded Firmware, RF Telemetry, 6-DoF Dynamics